Lab Task 1:

Translate the linear search algorithm into a program which either find the location LOC where ITEM appears in ARRAY or return LOC=0

Code:

```
#include <iostream>
using namespace std;

int main()
{
	const int arrSize = 10;
	int arr[arrSize] = { 21, 99, 150, 212, 320, 420, 510, 603, 834 , 911 };
	int item = 603;
	int loc = 0;

	for (int i = 0; i < arrSize; i++) {
		if (arr[i] == item) {
			loc = i + 1;
			break;
		}
	}

	if (loc != 0) {
		cout << "Item " << item << " found at location " << loc << endl;
	}
	else {
		cout << "Item not found in the array." << endl;
	}
}
```

Output:

```
Item 603 found at location 8
```

Lab Task 2:

Translate binary search and insertion algorithm into a program which finds either the location LOC where ITEM appears in ARRAY or the location LOC where ITEM should be inserted into ARRAY. (For object 3, 4, and 5 take an array of 10 elements given below)

Code:

```
#include <iostream>
using namespace std;

int main()
{
	const int ArrSize = 10;
	int arr[ArrSize] = { 3, 5, 8, 12, 15, 18, 20, 25, 30, 35 };
	int item;

	cout << "Enter item to search: ";
	cin >> item;

	int beg = 0;
	int end = ArrSize - 1;
	int loc = -1;

	while (beg <= end)
	{
		int mid = (beg + end) / 2;
		if (arr[mid] == item) {
			loc = mid + 1;
			break;
		}
		else if (item < arr[mid]) {
			end = mid - 1;
		}
		else {
			beg = mid + 1;
		}
	}

	if (loc != -1) {
		cout << "Item " << item << " found at location " << loc << endl;
	}
	else {
		cout << "Item not found in the array." << endl;
		cout << "Item should be inserted at location " << beg + 1 << endl;
	}
}
```

Output:

```
Enter item to search: 8
Item 8 found at location 3
```

Lab Task 3:

Write a program which uses Binary Search to search the elements52, and 33 and print it.

Code:

```
#include <iostream>
using namespace std;

int main() {
    const int n = 10;
    int arr[n] = { 11, 22, 33, 36, 45, 52, 57, 60, 64, 78 };

    int items[2] = { 52, 33 };

    for (int k = 0; k < 2; k++) {
        int item = items[k];
        int beg = 0;
        int end = n - 1;
        int loc = -1;

        while (beg <= end) {
            int mid = (beg + end) / 2;
            if (arr[mid] == item) {
                loc = mid;
                break;
            }
            else if (item < arr[mid]) {
                end = mid - 1;
            }
            else {
                beg = mid + 1;
            }
        }

        if (loc != -1) {
            cout << item << " found at index " << loc << endl;
        }
        else {
            cout << item << " not found" << endl;
        }
    }
}
```

Output:

```
52 found at index 5
33 found at index 2
```

Lab Task 4:

Write a program which uses Binary Search to search the elements 45 and 78 and delete it. NUMBERS= [11,22,33,36,45,52,57,60,64,78]

Code:

```
#include <iostream>
using namespace std;

int main()
{
    const int arrSIze = 10;
    int arr[arrSIze] = { 11, 22, 33, 36, 45, 52, 57, 60, 64, 78 };
    int n = arrSIze;
    int items[2] = { 45, 78 };

    for (int k = 0; k < 2; k++)
    {
        int item = items[k];
        int beg = 0, end = n - 1;
        int loc = -1;
        while (beg <= end)
        {
            int mid = (beg + end) / 2;

            if (arr[mid] == item)
            {
                loc = mid;
                break;
            }
            else if (item < arr[mid])
            {
                end = mid - 1;
            }
            else
            {
                beg = mid + 1;
            }
        }
        if (loc != -1)
        {
            for (int i = loc; i < n - 1; i++)
            {
                arr[i] = arr[i + 1];
            }
            n--;
        }
    }

    cout << "Updated Array: ";
    for (int i = 0; i < n; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}
```

Output:

```
Updated Array: 11 22 33 36 52 57 60 64
```